

Sarandeep balaji

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Profile

Enthusiastic **Electronics and Communication Engineering** graduate from Karpagam College of Engineering, specializing in **software development**. Skilled in **C and C++**, with a strong foundation in **object-oriented programming, data structures, and algorithms**. Experienced in **software design, debugging, and version control systems**. Passionate about developing **efficient and reliable applications**, and eager to contribute to **innovative software solutions** that address real-world challenges.

Education

2022 – 2026	Karpagam College Of Engineering,
Coimbatore, India	<i>Electronics And Communication Engineering</i>
	Cgpa upto 7th semester-(8.2)
2020 – 2022	Perks Matriculation Higher Secondary School
Coimbatore, India	<i>Karpagam College of Engineering</i>
	12th percentage -66%
	10th percentage -78.5%

Programing skills

c language

c++

python developer

Data structures and algorithms

DBMS

java developer

SOFT SKILLS

- Active Listening
- Communication Skills
- Teamwork
- Problem-Solving
- Critical Thinking

Core skills

Electronic devices

Electronic circuits

Digital Electronics

Microprocessor and Microcontroller

Computer Networks

Languages

- Tamil,English,Hindi

Projects

IoT-Based Pick and Place Robot Using ESP32

Designed and built a robotic system to pick up and place objects using sensors and actuators. The robot was controlled via an ESP32 microcontroller, with all axes managed through a web interface for remote control. This project aimed to automate repetitive tasks in industries like manufacturing and logistics.

Mountain Vehicle Collision Prevention Using Arduino

The Mountain Vehicle Collision Prevention System using ESP32 microcontroller helps prevent accidents on mountain roads by using sensors to detect oncoming vehicles around curves. When a vehicle is detected, the system alerts the driver with visual (LEDs) and audio (buzzer) signals, indicating potential danger and helping to avoid collisions.

Automatic Car Parking Allotment System Using Arduino

The Automatic Car Parking Allotment System using Arduino efficiently manages parking spaces by detecting available spots and guiding vehicles to them. Using ultrasonic sensors, the system detects free spaces and signals their availability via LEDs or an LCD display. When a car arrives, it is directed to the nearest

RFID Based Billing Counter Using 8051 Microcontroller

The RFID-Based Billing Counter using an 8051 Microcontroller automates the billing process in retail by using RFID technology to identify products with RFID tags. When an item with an RFID tag is scanned, the system retrieves and displays the product details and updates the bill on a 16x2 LCD screen, significantly speeding up and reducing errors in the checkout process.

IOT based landslide monitoring system using ESP32 and LORA

Developed a real-time landslide monitoring system using ESP32 and LoRa communication to detect early warning signs of soil movement. Integrated multiple sensors such as tilt (MPU6050), soil moisture, and vibration sensors to continuously monitor environmental conditions. The system processes sensor data and transmits it over long distances using LoRa for remote monitoring. Implemented threshold-based alert mechanisms to identify potential landslide risks and improve disaster preparedness. This project highlights skills in IoT, wireless communication, embedded systems, and environmental monitoring.

Airline Ticket Booking System

Developed a console-based Airline Ticket Booking System using Python and MySQL, applying Object-Oriented Programming (OOP) concepts such as classes, objects, and encapsulation. Designed modular components including User, Flight, Booking, and Ticket to handle core functionalities. Integrated MySQL database using mysql-connector to perform CRUD operations and manage relational data efficiently. Implemented features like booking confirmation, seat availability tracking, and ticket generation, ensuring proper data flow and persistence. Focused on structured, modular code design with basic error handling, making the system scalable and maintainable.

Certificates

Codechef

Codechef Certification On C Programming

Udemy

certificate completion in java training

Great Learning

certificate completion for python

Simplilearn

Certificate on Frontend development

Internships

01/2026 – 05/2026
bengaluru

Intership at Mphasis pvt ltd

Associate software developer

I completed a three-month internship at Mphasis Limited from January 16, 2026 to April 15, 2026, where I gained practical exposure to IT infrastructure and enterprise systems. During the internship, I worked on understanding the Gensler project and its real-world infrastructure workflows. I developed knowledge in networking concepts, including wireless networks, hubs, switches, and server architecture. I also gained hands-on experience with infrastructure management tools such as InfraGenie and explored cloud computing concepts using Microsoft Azure for resource management. This experience helped me build a strong foundation in system connectivity, deployment, and enterprise-level infrastructure operations.

12/2024 – 01/2025
Coimbatore, India

Internship at Auswag Info Control Company

During my internship at Auswag Info Control Company, I acquired hands-on experience in the design and development of robotic systems. I worked with advanced automation technologies to improve operational efficiency and precision. My responsibilities included integrating sensors and actuators, implementing navigation and object detection algorithms, and staying updated with emerging trends in robotics and automation. This experience significantly strengthened my technical foundation and equipped me with practical skills essential for my future career in the field.